

Elementary coupled systems

MG3416–Advanced Structural Acoustics - Lecture #2

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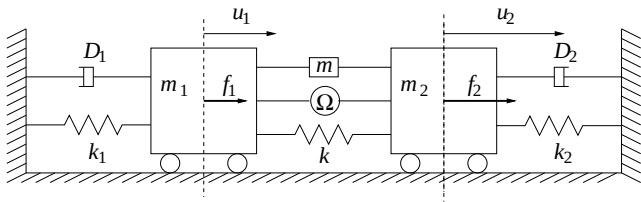
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Two-DOFs system

Notations and setting



- The two-degrees-of-freedom system:

$$\begin{bmatrix} M_1 & M \\ M & M_2 \end{bmatrix} \begin{pmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{pmatrix} + \begin{bmatrix} D_1 & \Omega \\ -\Omega & D_2 \end{bmatrix} \begin{pmatrix} \dot{u}_1 \\ \dot{u}_2 \end{pmatrix} + \begin{bmatrix} K_1 & K \\ K & K_2 \end{bmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}.$$

- Fundamental parameters for $\alpha \in \{1, 2\}$:

- $\omega_\alpha = \sqrt{\frac{K_\alpha}{M_\alpha}}$: the "blocked" natural (angular) frequency of the α^{th} oscillator,
- $\xi_\alpha = \frac{D_\alpha}{2\sqrt{K_\alpha M_\alpha}}$: the critical damping rate of the α^{th} oscillator,
- $\eta_\alpha = 2\xi_\alpha$: the loss factor of the α^{th} oscillator.

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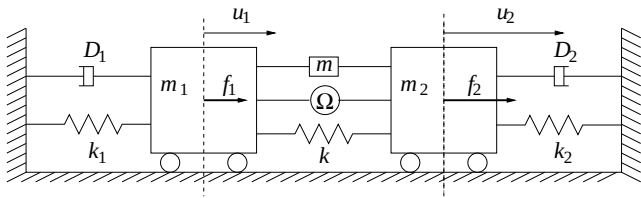
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- The two-degrees-of-freedom system:

$$\begin{bmatrix} M_1 & M \\ M & M_2 \end{bmatrix} \begin{pmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{pmatrix} + \begin{bmatrix} D_1 & \Omega \\ -\Omega & D_2 \end{bmatrix} \begin{pmatrix} \dot{u}_1 \\ \dot{u}_2 \end{pmatrix} + \begin{bmatrix} K_1 & K \\ K & K_2 \end{bmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}.$$

- Hypotheses:

- $K < \sqrt{K_1 K_2}$ and $M < \sqrt{M_1 M_2}$;
- Conservative gyroscopic coupling:

$$\left(\begin{bmatrix} 0 & \Omega \\ -\Omega & 0 \end{bmatrix} \mathbf{a}, \mathbf{a} \right) = 0, \quad \forall \mathbf{a} \in \mathbb{R}^2.$$

Energetic quantities

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Definition

- The *mechanical energy* of the two-DOFs system:

$$\mathcal{E}(t) = \mathcal{E}_1(t) + \mathcal{E}_2(t) + M\dot{u}_1(t)\dot{u}_2(t) + Ku_1(t)u_2(t),$$

where $\mathcal{E}_\alpha(t) = \frac{1}{2}M_\alpha\dot{u}_\alpha(t)^2 + \frac{1}{2}K_\alpha u_\alpha(t)^2$ for $\alpha \in \{1, 2\}$.

- The *dissipated power*: $\Pi_d(t) = \Pi_{d1}(t) + \Pi_{d2}(t)$, where $\Pi_{d\alpha}(t) = D_\alpha\dot{u}_\alpha(t)^2$ for $\alpha \in \{1, 2\}$.
- The *input power*: $\Pi_{IN}(t) = \Pi_{IN1}(t) + \Pi_{IN2}(t)$, where $\Pi_{IN\alpha}(t) = f_\alpha(t)\dot{u}_\alpha(t)$ for $\alpha \in \{1, 2\}$.

Energetic quantities

Power balance

- The powers exchanged by the oscillators:

$$\begin{aligned}\Pi_{12}(t) &= (\text{force } 1 \rightarrow 2) \times (\text{celerity } 2) \\ &= -(M\ddot{u}_1 - \Omega\dot{u}_1 + Ku_1)\dot{u}_2;\end{aligned}$$

$$\begin{aligned}\Pi_{21}(t) &= (\text{force } 2 \rightarrow 1) \times (\text{celerity } 1) \\ &= -(M\ddot{u}_2 + \Omega\dot{u}_2 + Ku_2)\dot{u}_1.\end{aligned}$$

- The instantaneous power of the two-DOFs system:

$$\Pi(t) = \dot{\mathcal{E}}(t) = \dot{\mathcal{E}}_1(t) + \dot{\mathcal{E}}_2(t) - \Pi_{12}(t) - \Pi_{21}(t),$$

- The instantaneous power balance reads:

$$\dot{\mathcal{E}}_1(t) + \dot{\mathcal{E}}_2(t) = \Pi_{\text{IN}}(t) - \Pi_{\text{d}}(t) + \Pi_{12}(t) + \Pi_{21}(t),$$

or for each oscillator:

$$\dot{\mathcal{E}}_{\alpha}(t) = \Pi_{\text{IN}\alpha}(t) - \Pi_{\text{d}\alpha}(t) + \Pi_{\beta\alpha}(t), \quad \beta \neq \alpha \in \{1, 2\}.$$

Stationary excitations

Wideband noise

- **Data:** $(F_{\alpha,t}, t \in \mathbb{R}), \alpha \in \{1, 2\}$, are $\mathbb{R}$ -valued second order, centered stochastic processes defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, and mean-square stationary.
- **Hypotheses:**

- $\exists \omega \mapsto S_{\alpha}(\omega) : \mathbb{R} \rightarrow \mathbb{R}_+ \text{ s.t.}$

$$S_{\alpha}(\omega) = S_{\alpha} \mathbb{1}_{I_{\alpha} \cup \underline{I}_{\alpha}}(\omega), \quad S_{\alpha} > 0,$$

where:

$$I_{\alpha} \cup \underline{I}_{\alpha} = \left[\omega_{\alpha} - \frac{\Delta\omega}{2}, \omega_{\alpha} + \frac{\Delta\omega}{2} \right] \cup \left[-\omega_{\alpha} - \frac{\Delta\omega}{2}, -\omega_{\alpha} + \frac{\Delta\omega}{2} \right];$$

- $(F_{1,t}, t \in \mathbb{R})$ and $(F_{2,t}, t \in \mathbb{R})$ are **uncorrelated**:

$$\mathbb{E}\{F_{1,t}F_{2,t'}\} = 0, \quad \forall(t, t') \in \mathbb{R} \times \mathbb{R}.$$

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Stationary excitations

Forced responses of the two-DOFs system

■ Remarks:

- $\omega \mapsto S_\alpha(\omega)$ is even and integrable;
- The auto-correlation function:

$$R_\alpha(t) = \int_{\mathbb{R}} e^{i\omega t} S_\alpha(\omega) d\omega = 2\Delta\omega S_\alpha \cos(\omega_\alpha t) \text{sinc}\left(\frac{\Delta\omega}{2}t\right)$$

is continuous, bounded, $R_\alpha(0) > 0$, $\lim_{|t| \rightarrow +\infty} R_\alpha(t) = 0$.

Proposition

- $(U_{\alpha,t}, t \in \mathbb{R})$, $\alpha \in \{1, 2\}$, are $\mathbb{R}$ -valued second order, centered stochastic processes defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, and mean-square stationary.
- The same holds for their mean-square derivatives $(\dot{U}_{\alpha,t}, t \in \mathbb{R})$ and $(\ddot{U}_{\alpha,t}, t \in \mathbb{R})$, $\alpha \in \{1, 2\}$.

Stationary excitations

Second-order properties of the forced responses

- The $\mathbb{C}^{2 \times 2}$ spectral density matrix of the forced response vector $\mathbf{U}_t^f = (U_{1,t}, U_{2,t})^\top$ reads:

$$\begin{aligned}\mathbf{S}_U(\omega) &= \hat{\mathbf{h}}_1(\omega) \otimes \overline{\hat{\mathbf{h}}_1(\omega)} S_1(\omega) + \hat{\mathbf{h}}_2(\omega) \otimes \overline{\hat{\mathbf{h}}_2(\omega)} S_2(\omega) \\ &= \frac{S_1(\omega)}{|D(\omega)|^2} \begin{bmatrix} |A_2(\omega)|^2 & -A_c(\omega)A_2(\omega) \\ -\overline{A_c(\omega)}A_2(\omega) & |A_c(\omega)|^2 \end{bmatrix} \\ &\quad + \frac{S_2(\omega)}{|D(\omega)|^2} \begin{bmatrix} |A_c(\omega)|^2 & -A_c(\omega)\overline{A_1(\omega)} \\ -\overline{A_c(\omega)}A_1(\omega) & |A_1(\omega)|^2 \end{bmatrix}\end{aligned}$$

where $D(\omega) = A_1(\omega)A_2(\omega) - |A_c(\omega)|^2$ and:

$$A_\alpha(\omega) = -\omega^2 M_\alpha + i\omega D_\alpha + K_\alpha,$$

$$A_c(\omega) = -\omega^2 M + i\omega\Omega + K.$$

- Besides:

$$\mathbb{E}\{U_{\alpha,t}\dot{U}_{\alpha,t}\} = \mathbb{E}\{\dot{U}_{\alpha,t}\ddot{U}_{\alpha,t}\} = 0, \quad \forall t \in \mathbb{R},$$

$$\mathbb{E}\{U_{1,t}\dot{U}_{2,t}\} = -\mathbb{E}\{\dot{U}_{1,t}U_{2,t}\},$$

$$\mathbb{E}\{\dot{U}_{1,t}\ddot{U}_{2,t}\} = -\mathbb{E}\{\ddot{U}_{1,t}\dot{U}_{2,t}\}.$$

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Second-order properties of the forced responses (extended proof)

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■ $(-\omega^2 \mathbf{M} + i\omega \mathbf{D} + \mathbf{K}) \hat{\mathbf{U}}^f = \hat{\mathbf{F}}$ thus (filtering):

$$\begin{aligned} \mathbf{S}_U(\omega) &= \hat{\mathbf{h}}(\omega) \mathbf{S}_F(\omega) \hat{\mathbf{h}}^*(\omega) \\ &= (-\omega^2 \mathbf{M} + i\omega \mathbf{D} + \mathbf{K})^{-1} \begin{bmatrix} S_1(\omega) & 0 \\ 0 & S_2(\omega) \end{bmatrix} \overline{(-\omega^2 \mathbf{M} + i\omega \mathbf{D} + \mathbf{K})}^{-\top} \\ &= \hat{\mathbf{h}}_1(\omega) \otimes \overline{\hat{\mathbf{h}}_1(\omega)} S_1(\omega) + \hat{\mathbf{h}}_2(\omega) \otimes \overline{\hat{\mathbf{h}}_2(\omega)} S_2(\omega) \\ &= \frac{S_1(\omega)}{|D(\omega)|^2} \begin{bmatrix} |A_2(\omega)|^2 & -A_c(\omega) A_2(\omega) \\ -\overline{A_c(\omega) A_2(\omega)} & |A_c(\omega)|^2 \end{bmatrix} \\ &\quad + \frac{S_2(\omega)}{|D(\omega)|^2} \begin{bmatrix} |A_c(\omega)|^2 & -A_c(\omega) \overline{A_1(\omega)} \\ -\overline{A_c(\omega) A_1(\omega)} & |A_1(\omega)|^2 \end{bmatrix} \end{aligned}$$

Stationary excitations

Second-order properties of the forced responses (extended proof)

- Mean-square derivation $\mathbf{R}_{\dot{U}U}(t') = \mathbb{E}\{\dot{U}_{t'+t} \otimes U_t\} = -\mathbf{R}_{U\dot{U}}(t')$, $\forall t, t' \in \mathbb{R}$, that is:

$$\begin{bmatrix} \mathbb{E}\{\dot{U}_{1,t'+t}U_{1,t}\} & \mathbb{E}\{\dot{U}_{1,t'+t}U_{2,t}\} \\ \mathbb{E}\{\dot{U}_{2,t'+t}U_{1,t}\} & \mathbb{E}\{\dot{U}_{2,t'+t}U_{2,t}\} \end{bmatrix} = - \begin{bmatrix} \mathbb{E}\{U_{1,t'+t}\dot{U}_{1,t}\} & \mathbb{E}\{U_{1,t'+t}\dot{U}_{2,t}\} \\ \mathbb{E}\{U_{2,t'+t}\dot{U}_{1,t}\} & \mathbb{E}\{U_{2,t'+t}\dot{U}_{2,t}\} \end{bmatrix}.$$

Hence taking $t' = 0$:

$$\mathbb{E}\{U_{\alpha,t}\dot{U}_{\alpha,t}\} = 0,$$

$$\mathbb{E}\{U_{1,t}\dot{U}_{2,t}\} = -\mathbb{E}\{\dot{U}_{1,t}U_{2,t}\} \quad \forall t \in \mathbb{R}.$$

- Likewise $\mathbf{R}_{\ddot{U}\dot{U}}(t') = \mathbb{E}\{\ddot{U}_{t'+t} \otimes \dot{U}_t\} = -\mathbf{R}_{\dot{U}\ddot{U}}(t')$, $\forall t, t' \in \mathbb{R}$, therefore taking $t' = 0$:

$$\mathbb{E}\{\dot{U}_{\alpha,t}\ddot{U}_{\alpha,t}\} = 0,$$

$$\mathbb{E}\{\dot{U}_{1,t}\ddot{U}_{2,t}\} = -\mathbb{E}\{\ddot{U}_{1,t}\dot{U}_{2,t}\} \quad \forall t \in \mathbb{R}.$$

Stationary excitations

Energetics of the stationary forced responses

- The instantaneous power balance reads:

$$\dot{\mathcal{E}}_{1,t} + \dot{\mathcal{E}}_{2,t} = \Pi_{\text{IN},t} - \Pi_{\text{d},t} + \Pi_{12,t} + \Pi_{21,t},$$

or for each oscillator:

$$\dot{\mathcal{E}}_{\alpha,t} = \Pi_{\text{IN}\alpha,t} - \Pi_{\text{d}\alpha,t} + \Pi_{\beta\alpha,t}, \quad \beta \neq \alpha \in \{1, 2\},$$

as equalities of second-order random variables.

- From the foregoing results, $\mathbb{E}\{\dot{\mathcal{E}}_{\alpha,t}\} = 0$ for both oscillators and $\mathbb{E}\{\Pi_{12,t}\} = -\mathbb{E}\{\Pi_{21,t}\}$; hence:

$$\begin{aligned}\mathbb{E}\{\Pi_{\text{IN},t}\} &= \mathbb{E}\{\Pi_{\text{d},t}\}, \\ \mathbb{E}\{\Pi_{\text{IN}\alpha,t}\} &= \mathbb{E}\{\Pi_{\text{d}\alpha,t}\} + \mathbb{E}\{\Pi_{\alpha\beta,t}\}.\end{aligned}$$

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■ Hypotheses (wideband excitation):

■ i $\xi_\alpha \ll 1;$

■ ii $\Delta\omega \gg b_\alpha = \pi\xi_\alpha\omega_\alpha$ for $\alpha \in \{1, 2\}.$

■ From the foregoing results:

$$\begin{aligned}\mathbb{E}\{\Pi_{12,t}\} &= \mathbb{E}\{-(M\ddot{U}_{1,t} - \Omega\dot{U}_{1,t} + KU_{1,t})\dot{U}_{2,t}\} \\ &= \Re \left\{ \int_{\mathbb{R}} i\omega \overline{A_c(\omega)} S_{U_1 U_2}(\omega) d\omega \right\} \\ &= \int_{\mathbb{R}} \frac{\omega^2 |A_c(\omega)|^2}{\pi |D(\omega)|^2} D_1 D_2 \left(\frac{\pi S_1(\omega)}{D_1} - \frac{\pi S_2(\omega)}{D_2} \right) d\omega \\ &= \left(\int_{\mathbb{R}} \frac{\omega^2 |A_c(\omega)|^2}{\pi |D(\omega)|^2} D_1 D_2 d\omega \right) \left(\mathbb{E}\{\mathcal{E}_{1,t}^b\} - \mathbb{E}\{\mathcal{E}_{2,t}^b\} \right),\end{aligned}$$

where $\mathbb{E}\{\mathcal{E}_{\alpha,t}^b\} = \frac{\pi S_\alpha}{D_\alpha}$: the average mechanical energy of the "blocked" α^{th} oscillator.

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■ Besides:

$$\begin{aligned}\mathbb{E}\{\mathcal{E}_{\alpha,t}\} &= \frac{1}{2}M_{\alpha}\mathbb{E}\{\dot{U}_{\alpha,t}^2\} + \frac{1}{2}K_{\alpha}\mathbb{E}\{U_{\alpha,t}^2\} \\ &= \frac{1}{2}M_{\alpha}\int_{\mathbb{R}}\omega^2 S_{U_{\alpha}U_{\alpha}}(\omega)\mathrm{d}\omega + \frac{1}{2}K_{\alpha}\int_{\mathbb{R}}S_{U_{\alpha}U_{\alpha}}(\omega)\mathrm{d}\omega \\ &\equiv A_{\alpha\alpha}S_{\alpha} + A_{\alpha\beta}S_{\beta}, \quad \beta \neq \alpha \in \{1,2\},\end{aligned}$$

from which one deduces S_1 and S_2 as functions of $\mathbb{E}\{\mathcal{E}_{1,t}\}$ and $\mathbb{E}\{\mathcal{E}_{2,t}\}$, and then $\mathbb{E}\{\Pi_{12,t}\}$ as a function of the latter:

$$\begin{aligned}\mathbb{E}\{\Pi_{12,t}\} &= \left(\int_{\mathbb{R}} \frac{\omega^2 |A_c(\omega)|^2}{|D(\omega)|^2 (A_{11}A_{22} - A_{12}A_{21})} \mathrm{d}\omega \right) \times \\ &\quad [(D_2A_{22} + D_1A_{21})\mathbb{E}\{\mathcal{E}_{1,t}\} - (D_1A_{11} + D_2A_{12})\mathbb{E}\{\mathcal{E}_{2,t}\}],\end{aligned}$$

or:

$$\boxed{\mathbb{E}\{\Pi_{12,t}\} = \omega_1\eta_{12}\mathbb{E}\{\mathcal{E}_{1,t}\} - \omega_2\eta_{21}\mathbb{E}\{\mathcal{E}_{2,t}\}}.$$

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- The average dissipated power:

$$\begin{aligned}\mathbb{E}\{\Pi_{d\alpha,t}\} &= D_\alpha \mathbb{E}\{\dot{U}_{\alpha,t}^2\} \\ &\simeq \omega_\alpha \eta_\alpha \mathbb{E}\{\mathcal{E}_{\alpha,t}\}.\end{aligned}$$

- The average power balance for each oscillator thus reads:

$$\boxed{\mathbb{E}\{\Pi_{IN\alpha,t}\} = \omega_\alpha \eta_\alpha \mathbb{E}\{\mathcal{E}_{\alpha,t}\} + \omega_\alpha \eta_{\alpha\beta} (\mathbb{E}\{\mathcal{E}_{\alpha,t}\} - \mathbb{E}\{\mathcal{E}_{\beta,t}\})},$$

owing to the [reciprocity relation](#):

$$\omega_\alpha \eta_{\alpha\beta} = \omega_\beta \eta_{\beta\alpha}, \quad \alpha \neq \beta \in \{1, 2\}.$$

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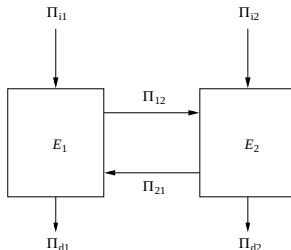
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$$\begin{bmatrix} \eta_1 + \eta_{12} & -\eta_{12} \\ -\eta_{21} & \eta_2 + \eta_{21} \end{bmatrix} \begin{pmatrix} \mathbb{E}\{\mathcal{E}_{1,t}\} \\ \mathbb{E}\{\mathcal{E}_{2,t}\} \end{pmatrix} = \begin{pmatrix} \frac{1}{\omega_1} \mathbb{E}\{\Pi_{IN1,t}\} \\ \frac{1}{\omega_2} \mathbb{E}\{\Pi_{IN2,t}\} \end{pmatrix}$$

- The data:
 - η_α : the **loss factor**,
 - $\eta_{\alpha\beta}$: the **coupling loss factor**,
 - $\mathbb{E}\{\Pi_{IN\alpha,t}\}$: the power input.
- The unknowns: $\mathbb{E}\{\mathcal{E}_{\alpha,t}\}$, $\alpha \in \{1, 2\}$.

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- **Example #1:** if $\eta_{\alpha\beta} \ll \eta_\alpha$: **weak coupling**,

$$\mathbb{E}\{\mathcal{E}_{\alpha,t}\} \simeq \frac{\mathbb{E}\{\Pi_{\text{IN}\alpha,t}\}}{\omega_\alpha \eta_\alpha} = \mathbb{E}\{\mathcal{E}_{\alpha,t}^b\}.$$

- **Example #2:** if $\eta_{\alpha\beta} \gg \eta_\alpha$: **strong coupling**,

$$\mathbb{E}\{\mathcal{E}_{1,t}\} \simeq \mathbb{E}\{\mathcal{E}_{2,t}\} \simeq \frac{\underline{\Pi}_{\text{IN}}}{\underline{\Delta}},$$

where $\underline{\Pi}_{\text{IN}} = \frac{1}{2}(\mathbb{E}\{\Pi_{\text{IN}1,t}\} + \mathbb{E}\{\Pi_{\text{IN}2,t}\})$ and $\underline{\Delta} = \frac{1}{2}(\omega_1 \eta_1 + \omega_2 \eta_2)$.

- **Remarks:**

- η_{12} and η_{21} depend on η_1 and η_2 ;
- if $\frac{M}{\sqrt{M_1 M_2}} \simeq \frac{K}{\sqrt{K_1 K_2}} \simeq \frac{\Omega}{\sqrt{D_1 D_2}} = O(\varepsilon)$, then $\eta_{\alpha\beta} = O(\varepsilon^2)$.

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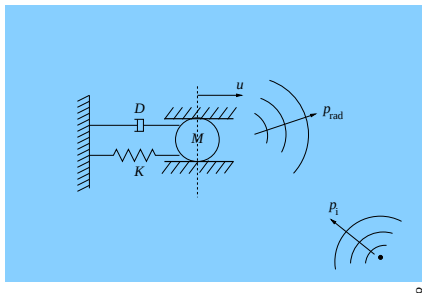
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- The oscillator ($\omega_p = \sqrt{\frac{K}{M}}$, $\eta_p = \frac{D}{\sqrt{KM}}$):

$$\begin{cases} M\ddot{u}(t) + D\dot{u}(t) + Ku(t) = Sp(t), & t \in \mathbb{R}, \\ u(0) = u_0, \\ \dot{u}(0) = v_0; \end{cases}$$

- S : surface seen by the fluid, p : fluid pressure.

Oscillator coupled to an acoustic fluid

Notations and setting

- The acoustic fluid: homogeneous, compressible, inviscid, gravity effects are neglected.
- Irrotational motion $\nabla \times \mathbf{v} = \mathbf{0}$, s.t. the fluid velocity $\mathbf{v}$ reads $\mathbf{v} = \nabla \psi$ and the fluid pressure reads $p = -\varrho_f \partial_t \psi$, where the **velocity potential** ψ satisfies:

$$\left\{ \begin{array}{l} \frac{1}{c_f^2} \frac{\partial^2 \psi}{\partial t^2} - \Delta \psi = g_{\text{IN}}(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega_f \times \mathbb{R}, \\ \frac{\partial \psi}{\partial \mathbf{n}}(\mathbf{x}_S, t) \stackrel{\text{def}}{=} \mathbf{v}(\mathbf{x}_S, t) \cdot \mathbf{n} = \dot{u}(t); \end{array} \right.$$

ϱ_f : fluid density, c_f : sound speed, $\mathbf{x}_S$: position of the oscillator at rest.

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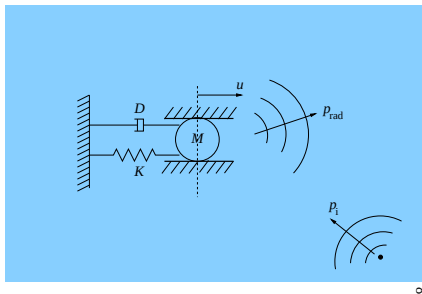
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- The velocity potential reads:

$$\psi(\mathbf{x}, t) = \psi_{\text{IN}}(\mathbf{x}, t) + \psi_{\text{d0}}(\mathbf{x}, t) + \psi_{\text{rad}}(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega_f \times \mathbb{R},$$

where:

- ψ_{IN} : incident potential (data);
- ψ_{d0} : diffracted potential, the oscillator being fixed;
- ψ_{rad} : radiated potential.

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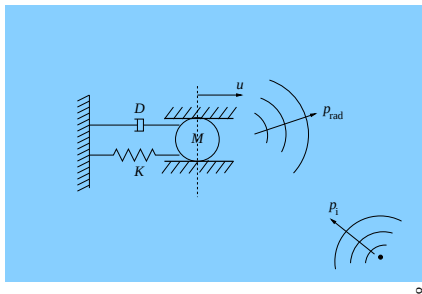
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- The **diffracted potential** ψ_{d0} satisfies:

$$\begin{cases} \frac{1}{c_f^2} \frac{\partial^2 \psi_{d0}}{\partial t^2} - \Delta \psi_{d0} = 0, & (\mathbf{x}, t) \in \Omega_f \times \mathbb{R}, \\ \frac{\partial \psi_{d0}}{\partial \mathbf{n}}(\mathbf{x}_S, t) = -\frac{\partial \psi_{IN}}{\partial \mathbf{n}}(\mathbf{x}_S, t). \end{cases}$$

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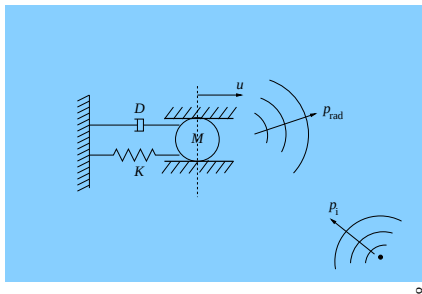
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- The **radiated potential** ψ_{rad} satisfies:

$$\begin{cases} \frac{1}{c_f^2} \frac{\partial^2 \psi_{\text{rad}}}{\partial t^2} - \Delta \psi_{\text{rad}} = 0, & (\mathbf{x}, t) \in \Omega_f \times \mathbb{R}, \\ \frac{\partial \psi_{\text{rad}}}{\partial \mathbf{n}}(\mathbf{x}_S, t) = \dot{u}(t). \end{cases}$$

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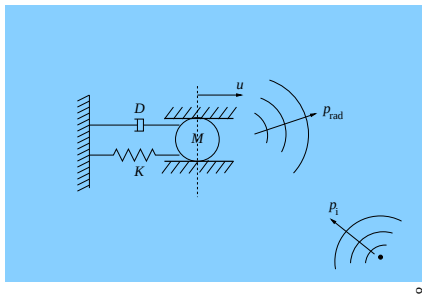
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- The incident potential ψ_{IN} satisfies:

$$\frac{1}{c_f^2} \frac{\partial^2 \psi_{\text{IN}}}{\partial t^2} - \Delta \psi_{\text{IN}} = g_{\text{IN}}(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \mathbb{R}^3 \times \mathbb{R},$$

for some sound source g_{IN} in the full physical space.

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- The exterior Helmholtz problem:

$$\left\{ \begin{array}{ll} \Delta \psi + k_f^2 \psi = 0 & \text{in } \Omega_f, \\ \frac{\partial \psi}{\partial \mathbf{n}} = v & \text{on } \partial \Omega_f, \\ |\psi| = O\left(\frac{1}{r}\right), \quad \left| \frac{\partial \psi}{\partial r} + i k_f \psi \right| = O\left(\frac{1}{r^2}\right) & \text{as } r = \|\mathbf{x}\| \rightarrow +\infty, \end{array} \right.$$

where $k_f = \frac{\omega}{c_f}$, and $\partial \Omega_f = \Gamma$ is the interface between the fluid and the structure—the oscillator.

- Sommerfeld radiation conditions ”at infinity”: the radiated waves are almost plane and do not propagate toward Γ .

Solving the fluid equations

Boundary impedance

- The exterior Helmholtz problem admits a unique solution $\forall k_f \in \mathbb{R}$, hence:

- There exists a linear operator $\mathcal{B}_\Gamma(k_f) : \mathcal{C}'_\Gamma \rightarrow \mathcal{C}_\Gamma$ s.t.:

$$\psi|_\Gamma(\omega) = \mathcal{B}_\Gamma(k_f)v, \quad \text{on } \Gamma;$$

- There exists a linear operator $\mathcal{R}_x(k_f) : \mathcal{C}'_\Gamma \rightarrow \mathbb{C}$ s.t.:

$$\psi(x, \omega) = \mathcal{R}_x(k_f)v, \quad x \in \Omega_f,$$

with $\mathcal{C}_\Gamma$: the set of admissible fields on Γ ($\mathcal{C}'_\Gamma$: its dual).

- $Z_\Gamma(\omega) = -i\omega\rho_f S \mathcal{B}_\Gamma(k_f)$ is the **acoustic impedance boundary operator**, with $S = |\Gamma|$;
- $\mathcal{Z}_x(\omega) = -i\omega\rho_f S \mathcal{R}_x(k_f)$ is the **radiation impedance operator**, with $S = |\Gamma|$.

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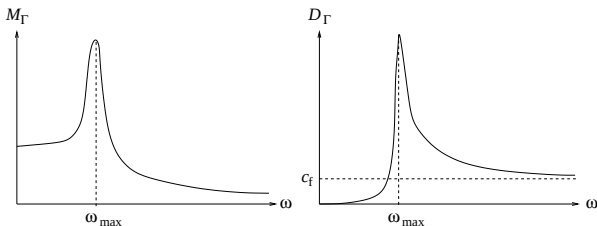
Boundary impedance

- The boundary impedance $\mathcal{B}_\Gamma(\frac{\omega}{c_f})$ is symmetric and reads:

$$-\omega^2 \mathcal{B}_\Gamma \left(\frac{\omega}{c_f} \right) = -\omega^2 M_\Gamma \left(\frac{\omega}{c_f} \right) + i\omega D_\Gamma \left(\frac{\omega}{c_f} \right),$$

where:

- The **reactive part** $\omega \mapsto M_\Gamma(\frac{\omega}{c_f})$ (left) is generally unsigned, though it is positive if $\mathbb{R}^3 \setminus \bar{\Omega}_f$ is convex;
- The **resistive part** $\omega \mapsto D_\Gamma(\frac{\omega}{c_f})$ (right) is positive.



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- Diffracted velocity potential and fluid pressure:

$$\begin{aligned}\hat{\psi}_{\text{d0}}(\omega)|_{\Gamma} &= -\mathcal{B}_{\Gamma}(k_f) \frac{\partial \hat{\psi}_{\text{IN}}}{\partial \mathbf{n}} & \text{and} & & \hat{p}_{\text{d0}}(\omega)|_{\Gamma} &= -\mathcal{B}_{\Gamma}(k_f) \frac{\partial \hat{p}_{\text{IN}}}{\partial \mathbf{n}} \\ \hat{\psi}_{\text{d0}}(\mathbf{x}, \omega) &= -\mathcal{R}_{\mathbf{x}}(\omega) \frac{\partial \hat{\psi}_{\text{IN}}}{\partial \mathbf{n}} & & & \hat{p}_{\text{d0}}(\mathbf{x}, \omega) &= -\mathcal{R}_{\mathbf{x}}(\omega) \frac{\partial \hat{p}_{\text{IN}}}{\partial \mathbf{n}}, \quad \mathbf{x} \in \Omega_f.\end{aligned}$$

- Radiated velocity potential and fluid pressure:

$$\begin{aligned}\hat{\psi}_{\text{rad}}(\omega)|_{\Gamma} &= i\omega \mathcal{B}_{\Gamma}(k_f) \hat{u}(\omega) & \text{and} & & S\hat{p}_{\text{rad}}(\omega)|_{\Gamma} &= i\omega Z_{\Gamma}(k_f) \hat{u}(\omega) \\ \hat{\psi}_{\text{rad}}(\mathbf{x}, \omega) &= i\omega \mathcal{R}_{\mathbf{x}}(\omega) \hat{u}(\omega) & & & S\hat{p}_{\text{rad}}(\mathbf{x}, \omega) &= i\omega Z_{\mathbf{x}}(\omega) \hat{u}(\omega), \quad \mathbf{x} \in \Omega_f.\end{aligned}$$

- The (**linear**) scattering operator $\mathcal{T}_{\mathbf{x}}(\omega)$ s.t.:

$$\hat{p}_{\text{IN}}(\mathbf{x}, \omega) + \hat{p}_{\text{d0}}(\mathbf{x}, \omega) = \mathcal{T}_{\mathbf{x}}(\omega) \hat{p}_{\text{IN}}(\omega)|_{\Gamma}, \quad \mathbf{x} \in \Omega_f.$$

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Frequency response function

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- Equation of motion of the oscillator in the frequency domain:

$$\begin{aligned}(-\omega^2 M + i\omega D + K)\hat{u}(\omega) &= S(\hat{p}_{\text{IN}}(\omega) + \hat{p}_{\text{d0}}(\omega) + \hat{p}_{\text{rad}}(\omega)) + \hat{f}_e(\omega) \\ &= S\mathcal{T}_{\mathbf{x}_S}(\omega)\hat{p}_{\text{IN}}(\omega) + \omega^2 \varrho_f S\mathcal{B}_{\mathbf{x}_S}(k_f)\hat{u}(\omega) + \hat{f}_e(\omega),\end{aligned}$$

where $\hat{f}_e(\omega) = Du_0 + M(i\omega u_0 + v_0)$ is an equivalent load accounting for the initial conditions.

- Therefore, the frequency response function of the oscillator coupled to the fluid reads:

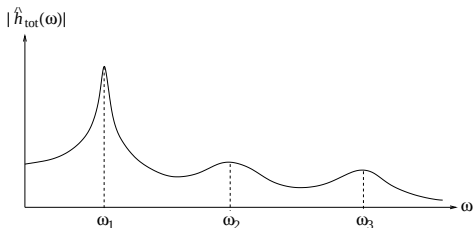
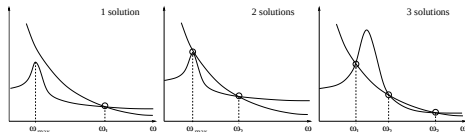
$$\hat{\mathbf{h}}_{\text{tot}}(\omega) = \left[-\omega^2 \left(M + \varrho_f S M_{\Gamma} \left(\frac{\omega}{c_f} \right) \right) + i\omega \left(D + \varrho_f S D_{\Gamma} \left(\frac{\omega}{c_f} \right) \right) + K \right]^{-1}.$$

Oscillator coupled to an acoustic fluid

Wet eigenfrequencies

- The "wet" eigenfrequencies of the oscillator coupled to the fluid are the solutions of:

$$\left[1 + \frac{\rho_f S}{M} M_\Gamma \left(\frac{\omega}{c_f} \right) \right] \omega^2 = \omega_p^2.$$



Stationary incident pressure

Forced response of the SDOF oscillator

- **Data:** $(P_t, t \in \mathbb{R})$ is a $\mathbb{R}$ -valued second order, centered stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, and mean-square (m.s.) stationary.
- **Hypothesis:** $\exists \omega \mapsto S_P(\omega) : \mathbb{R} \rightarrow \mathbb{R}_+$ s.t.

$$S_P(\omega) = S_0 \mathbb{1}_{I_0 \cup \underline{I}_0}(\omega), \quad S_0 > 0,$$

where:

$$I_0 \cup \underline{I}_0 = \left[\omega_0 - \frac{\Delta\omega}{2}, \omega_0 + \frac{\Delta\omega}{2} \right] \cup \left[-\omega_0 - \frac{\Delta\omega}{2}, -\omega_0 + \frac{\Delta\omega}{2} \right].$$

Proposition (filtering)

- $(U_t^f, t \in \mathbb{R})$ is a $\mathbb{R}$ -valued second order stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, centered and m.s. stationary s.t. $S_U(\omega) = |\hat{\mathbf{h}}_{\text{tot}}(\omega) S \mathcal{T}_{\mathbf{x}_S}(\omega)|^2 S_P(\omega)$.

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Forced response of the SDOF oscillator

- **Data:** $(P_t, t \in \mathbb{R})$ is a $\mathbb{R}$ -valued second order, centered stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, and mean-square (m.s.) stationary.
- **Hypothesis:** $\exists \omega \mapsto S_P(\omega) : \mathbb{R} \rightarrow \mathbb{R}_+$ s.t.

$$S_P(\omega) = S_0 \mathbb{1}_{I_0 \cup \underline{I}_0}(\omega), \quad S_0 > 0,$$

where:

$$I_0 \cup \underline{I}_0 = \left[\omega_0 - \frac{\Delta\omega}{2}, \omega_0 + \frac{\Delta\omega}{2} \right] \cup \left[-\omega_0 - \frac{\Delta\omega}{2}, -\omega_0 + \frac{\Delta\omega}{2} \right].$$

Proposition (m.s. derivation)

- *The same holds for its m.s. derivatives $(\dot{U}_t^f, t \in \mathbb{R})$ and $(\ddot{U}_t^f, t \in \mathbb{R})$, with $S_{\dot{U}}(\omega) = \omega^2 S_U(\omega)$, $S_{\ddot{U}}(\omega) = \omega^4 S_U(\omega)$, respectively.*

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Energetics of the stationary forced response

- The average mechanical energy of the oscillator:

$$\begin{aligned}\mathbb{E}\{\mathcal{E}_t\} &= \frac{1}{2}M \int_{\mathbb{R}} \omega^2 \left(1 + \frac{\varrho_f S}{M} M_{\Gamma} \left(\frac{\omega}{c_f} \right) \right) S_U(\omega) d\omega \\ &\quad + \frac{1}{2}K \int_{\mathbb{R}} S_U(\omega) d\omega \\ &\stackrel{\text{def}}{=} (M + M_{\text{rad}}(\omega_0)) \int_{\mathbb{R}} \omega^2 S_U(\omega) d\omega ,\end{aligned}$$

where $M_{\text{rad}}(\omega_0)$ is an equivalent added mass.

- The average **radiated power**:

$$\begin{aligned}\mathbb{E}\{\Pi_{\text{rad},t}\} &= \int_{\mathbb{R}} \varrho_f S D_{\Gamma} \left(\frac{\omega}{c_f} \right) \omega^2 S_U(\omega) d\omega \\ &\stackrel{\text{def}}{=} \omega_0 \eta_{\text{rad}}(\omega_0) \mathbb{E}\{\mathcal{E}_t\} ,\end{aligned}$$

where $\eta_{\text{rad}}(\omega_0)$ is an equivalent added loss factor.

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- The average dissipated power thus reads:

$$\begin{aligned}\mathbb{E}\{\Pi_{d,t}\} &= D\mathbb{E}\{\dot{U}_t^2\} \\ &= M\omega_p\eta_p \int_{\mathbb{R}} \omega^2 S_U(\omega) d\omega \\ &= \frac{M}{M + M_{\text{rad}}(\omega_0)} \omega_p\eta_p \mathbb{E}\{\mathcal{E}_t\} .\end{aligned}$$

- The average input power:

$$\mathbb{E}\{\Pi_{\text{IN},t}\} = \Re \int_{\mathbb{R}} i\omega \hat{\mathbf{h}}_{\text{tot}}(\omega) |S\mathcal{T}_{\mathbf{x}_S}(\omega)|^2 S_P(\omega) d\omega .$$

Stationary incident pressure

Power balance for the stationary forced response

- The instantaneous power balance reads:

$$\dot{\mathcal{E}}_t = \Pi_{\text{IN},t} - \Pi_{\text{d},t} - \Pi_{\text{rad},t},$$

as an equality of second-order random variables.

- Considering the mathematical expectation with $\mathbb{E}\{\mathcal{E}_t\} = \text{Constant}$ and $\mathbb{E}\{\dot{\mathcal{E}}_t\} = 0$:

$$\begin{aligned}\mathbb{E}\{\Pi_{\text{IN},t}\} &= \mathbb{E}\{\Pi_{\text{d},t}\} + \mathbb{E}\{\Pi_{\text{rad},t}\} \\ &= \omega_0 \eta_{\text{tot}}(\omega_0) \mathbb{E}\{\mathcal{E}_t\},\end{aligned}$$

where:

$$\begin{aligned}\eta_{\text{tot}}(\omega_0) &= \eta_{\text{rad}}(\omega_0) + \frac{\omega_p}{\omega_0} \left(\frac{M}{M + M_{\text{rad}}(\omega_0)} \right) \eta_p \\ &\simeq \eta_{\text{rad}}(\omega_0) + \eta_p \sqrt{\frac{M}{M + \varrho_f S M_\Gamma(\frac{\omega_0}{c_f})}}\end{aligned}$$

if ω_0 is close to a "wet" eigenfrequency.

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- The power flow between two coupled oscillators is proportional to the difference of their mechanical energies, if:
 - 1 The oscillators are weakly dissipative;
 - 2 Their coupling is conservative;
 - 3 They are excited by uncorrelated wideband noises, the bandwidths of which are large with respect to the equivalent bandwidths of the oscillators.
- "Equivalence" with the time average energetic quantities for the forced response of the randomized two-DOFs system to harmonic excitations.
- The average radiated power of the oscillator coupled to an acoustic fluid is proportional to its mechanical energy. This effect can be simply characterized by an equivalent added loss factor in the energy balance.
- **Outlook:** continuous systems.

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